IIBS IT Infrastructure Guide

Dual-Router Network Configuration, DHCP Subnetting & Redundancy Standard

1. Network Topology & Overview

The IIBS campus network connects **180+ active client computers** across multiple interconnected and department switches. The ISP provides an enterprise Demarcation Switch configured with **two physical uplink ports** (Dual Static Public IPs). To ensure 100% network uptime, prevent DHCP conflicts, and eliminate single-point-of-failure outages, the network is segmented across two high-performance TP-Link Archer C6 routers.

Router Unit	ISP Uplink (WAN)	Internal LAN Subnet	Client Scope
Router 1 (Main)	ISP Port 1 Static IP: **103.39.127.131**	IP: **192.168.0.1** Mask: 255.255.255.0	Section A / Lab 1 / Staff (~60 PCs)
Router 2 (Secondary)	ISP Port 2 Static IP: **103.39.127.133**	IP: **192.168.1.1** Mask: 255.255.255.0	Section B / Lab 2 / Students (~120 PCs)

2. Router 1 Configuration Guide (Section A)

Setting Parameter	Configuration Value	Description & Notes
WAN Connection	Static IP (ISP Port 1)	IP: 103.39.127.131
LAN IP Address	**192.168.0.1**	Router 1 Management & Gateway
Subnet Mask	**255.255.255.0**	Standard Class C /24 Subnet
DHCP Server	**■■ Enabled**	Issues IP to Section A devices
IP Address Pool	**192.168.0.50 — 192.168.0.250**	Leaves .2 to .49 for static devices
Address Lease Time	**120** minutes	Optimal lease turnover time
Default Gateway	**192.168.0.1**	Points to Router 1
Primary DNS	**1.1.1.2** (or 8.8.8.8)	Cloudflare Security DNS
Secondary DNS	**9.9.9.9** (or 1.1.1.1)	Quad9 Malware Filtering DNS

3. Router 2 Configuration Guide (Section B)

Setting Parameter	Configuration Value	Description & Notes
WAN Connection	Static IP (ISP Port 2)	IP: 103.39.127.133
LAN IP Address	**192.168.1.1**	Router 2 Management & Gateway
Subnet Mask	**255.255.255.0**	Independent /24 Subnet
DHCP Server	**■■ Enabled**	Issues IP to Section B devices
IP Address Pool	**192.168.1.50 — 192.168.1.250**	Leaves .2 to .49 for static devices
Address Lease Time	**120** minutes	Optimal lease turnover time

Default Gateway	**192.168.1.1**	Points to Router 2 (Self)
Primary DNS	**1.1.1.2** (or 8.8.8.8)	Cloudflare Security DNS
Secondary DNS	**9.9.9.9** (or 1.1.1.1)	Quad9 Malware Filtering DNS

4. Emergency ISP Outage Failover Procedures

If either ISP Public IP undergoes an outage or line fault, follow these standard operating procedures to restore internet connectivity:

Outage Scenario	Impacted Section	Immediate Action Required
ISP Line 1 Down (103.39.127.131)	Section A (192.168.0.x)	1. Log into Router 1 (192.168.0.1). 2. Go to **Advanced $\rightarrow$ Network $\rightarrow$ Internet**. 3. Switch WAN Static IP to 103.39.127.133 OR change Gateway in DHCP to 192.168.1.1.
ISP Line 2 Down (103.39.127.133)	Section B (192.168.1.x)	1. Log into Router 2 (192.168.1.1). 2. Go to **Advanced $\rightarrow$ Network $\rightarrow$ Internet**. 3. Switch WAN Static IP to 103.39.127.131 OR change Gateway in DHCP to 192.168.0.1.
Single Router Hardware Failure	Failed Unit's Section	Connect the distribution switch of the failed router directly into one of the open LAN ports (Yellow) of the operational router.

5. Router Hardening & Performance Optimization

To ensure continuous high-speed routing across 180+ computers without NAT memory bottlenecks or security vulnerabilities:

• **Automated Reboot Schedule:** In both routers, navigate to *Advanced $\rightarrow$ System Tools $\rightarrow$ Reboot Schedule* and enable daily reboots at **03:00 AM**. This purges stale NAT session tables.

• **UPnP Disabled:** In *Advanced $\rightarrow$ NAT Forwarding $\rightarrow$ UPnP*, set UPnP to **OFF** to prevent unauthorized automatic port forwarding from infected endpoints.

• **Disable Remote WAN Management:** Ensure *Advanced $\rightarrow$ System Tools $\rightarrow$ Administration $\rightarrow$ Remote Management* is **Disabled** so the router web console is never exposed to the public internet.

• **Encrypted Wi-Fi:** Ensure wireless security uses **WPA2-PSK (AES)** or WPA3 with strong passphrase protection, with WPS turned off.

6. Web Application Security Reference

The institutional ticketing and asset management web application deployed at https://iibsservicerequest.vercel.app is secured with:

• **Cryptographic HMAC-SHA256 Token Authentication:** Administrative routes require verified session tokens.

• **Protected Static Assets:** Spreadsheets (.xlsx), documents (.docx), and official director signature assets are blocked from direct public HTTP download.

• **HTTP Edge Security Headers:** Enforced X-Content-Type-Options: nosniff, X-Frame-Options: SAMEORIGIN, and strict referrer policies.